

STAT1050 (2025/26)	Statistical Methods for Time Series	Contribution: 100% of course
Course Leader: Dr Konstantinos Skindilias	Time Series Analysis Coursework	Deadline Date: 3rd December 2024
This coursework will be marked anonymously YOU MUST NOT PUT ANY INDICATION OF YOUR IDENTITY IN YOUR SUBMISSION		
This coursework should take an average student who is up-to-date with tutorial work approximately 15 hours Feedback and grades are normally made available within 15 working days of the coursework deadline		
Learning Outcomes: 1. Critically evaluate the generated model output and underlying model assumptions together with their implications of respective advanced modelling and statistical techniques for time series analysis. 2. Demonstrate critical awareness by making a result driven choice of the most appropriate statistical tools to analyse time series data, model fitting/ estimation, and forecasting. Apply different linear regression models to time series data, select challenger models and critically assess when to use Machine Learning libraries. 3. Demonstrate a comprehensive understanding in autoregressive (AR) and moving average (MA) processes, their coupling as ARIMA integrated processes by applying them to a range of time series data using a suitable statistical package and where applicable Machine Learning techniques. Critically evaluate and interpret program output and be able to conduct a critical report for discussing the results and implications of the study.		

Plagiarism is presenting somebody else's work as your own. It includes copying information directly from the Web or books without referencing the material; submitting joint coursework as an individual effort; copying another student's coursework; stealing coursework from another student and submitting it as your own work. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set down by the University. Please see your student handbook for further details of what is / isn't plagiarism.

All material copied or amended from any source (e.g. internet, books) must be referenced correctly according to the reference style you are using.

Your work will be submitted for plagiarism checking. Any attempt to bypass our plagiarism detection systems will be treated as a severe Assessment Offence.

Coursework Submission Requirements

- **An electronic copy of your work for this coursework must be fully uploaded on the Deadline Date of **3rd December 2024 (+2 days grace period)** using the link on the coursework Moodle page for STAT1050.**
- **For this coursework you must submit a single PDF document. In general, any text in the document must not be an image (i.e. must not be scanned) and would normally be generated from other documents (e.g. MS Office using "Save As .. PDF"). An exception to this is handwritten mathematical notation, but when scanning does ensure the file size is not excessive.**
- **There are limits on the file size (see the relevant course Moodle page).**
- **Make sure that any files you upload are virus-free and not protected by a password or corrupted otherwise they will be treated as null submissions.**
- **Your work will not be printed in colour. Please ensure that any pages with colour are acceptable when printed in Black and White.**
- **You must NOT submit a paper copy of this coursework.**
- **All coursework must be submitted as above. Under no circumstances can they be accepted by academic staff**

The University website has details of the current Coursework Regulations, including details of penalties for late submission, procedures for Extenuating Circumstances, and penalties for Assessment Offences. See <http://www2.gre.ac.uk/current-students/regs>

• Detailed Specification

Coursework must be completed individually. Students are allowed to work in groups in order to produce results and discuss implications of their findings

• Deliverables

Single PDF document. No Excel file should be uploaded and R output.

• Grading Criteria

• Marks will be distributed as follows:	
0-30	An unsatisfactory submission that demonstrates little understanding of the subject matter.
31-39	A substandard submission which has no critical appraisal of the numerical methods and concepts of scientific methodology.
40-49	A satisfactory submission with a basic understanding of the models used and basic descriptions with no proofs
50-59	A submission which includes a reasonable understanding of the underlying models, has some proofs and basic discussion.
60-69	A good submission which includes a good understanding of the numerical methods and concepts of scientific algorithms. Elaboration and discussion of the underlying theory, implications of tests, results and underlying models.
70+	A very good submission that clearly demonstrates the relative merits of the underlying models and implications of their use. Detailed discussion of the results, underlying theory and relating

	the dynamics of the underlying time series to those implied by the models. Equations and proofs are given where necessary with critical derivation and discussion.
80+	An excellent submission which demonstrates a clear understanding of the underlying models and implications of their use. Clear derivation and presentation of all necessary formulas used and implications of their dynamics relevant to the dynamics of the time series under investigation. Excellent discussion of the underlying theory of discrete time stochastic processes, methodology and theory of financial markets.

Coursework: ARIMA Model Fitting and Analysis

STAT-1050 Statistical Methods for Time Series

Deadline: Upload your report to Moodle by **3rd December 2025 (23:30 UK time)**

Submission Format: PDF processed report, incorporating tables and plots derived from R output. Code chunks should be converted into readable tables or visuals rather than raw script. **Do not submit R files or raw code.**

Instructions:

- **PDF Requirements:** Your report should include clear tables, plots, and commentary for readability and clarity.
- **Equations and Symbols:** Use the equation editor to format all mathematical expressions and symbols.
- **Length Restriction:** Limit the report to **4 pages**, including all visuals and commentary.
- **Objective:** Analyze and fit an ARIMA model to the **S&P 500 index data** covering the period from **2010 to 2025**.
- **No Screenshots or Raw Code Output:** Do not include screenshots or raw code output. Provide **clear code syntax for specific commands responsible for generating each result** without displaying entire code blocks.
- **Mathematical Representation:** Show all relevant mathematical equations and derivations where necessary to explain each step, particularly those related to ARIMA modeling.
- **Hypothesis Testing:** For each hypothesis test, clearly define the **null and alternative hypotheses** and interpret p-values in the context of these hypotheses.

Assessment Criteria:

This coursework evaluates your **understanding, mathematical reasoning, and critical thinking skills** in ARIMA model fitting and time series analysis. You will be assessed based on the following:

- **Reference to Lecture Material and Resources:**
 - Apply the methods, tests, and concepts discussed in lectures. Ensure that each analysis step reflects a well-rounded understanding of ARIMA modeling concepts, with appropriate mathematical rigor.
- **Demonstrate Critical Thinking and Mathematical Justification:**
 - **Justify Each Test and Approach with Mathematics:** Explain why you choose particular methods, tests, or models in your analysis, relating them to the characteristics of your data. Use mathematical formulations to support your choice of modeling approaches.
 - **Mathematical Representation of Hypothesis Testing:** Clearly explain each hypothesis test, specifying the null and alternative hypotheses with

the relevant mathematical notation. Discuss p-values and what they imply about the data, referencing equations when possible.

- **Depth of Interpretation and Analysis:** Go beyond a surface-level explanation of your results. Assessments will focus on how you analyze, interpret, and explain the implications of your findings for the data and model quality. Mathematical justification should support your interpretations wherever applicable.
 - **Model Selection and Evaluation:**
 - **Assess Model Quality Mathematically:** Discuss the strengths and limitations of each model, using mathematical evidence where applicable, and explain why certain models are favored over others based on observed performance and fit quality.
 - **Justify Parameter Choices:** Justify the choice of parameters and model specifications, showing an understanding of trade-offs between model complexity and fit quality, with relevant mathematical explanations.
 - **Compare Approaches with Critical and Mathematical Insight:** Critically analyze different modeling approaches or configurations, and discuss the reasons behind selecting one over the other, providing mathematical reasoning wherever possible.
 - **Clarity, Structure, and Presentation:**
 - Ensure the report is well-structured and that your arguments are supported with clear visuals, tables, and commentary, making the report accessible within the page limit.
 - Use mathematical notation, formatting, presentation, and clear language to communicate professionally.
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COURSEWORK Questions and marks

1. **Data Retrieval and Preprocessing** (15 marks)
2. **Preliminary Analysis and Model Identification** (20 marks)
3. **ARIMA Model Fitting and Comparison** (30 marks)
4. **Residual Diagnostics and Assumption Validation** (20 marks)
5. **Model Evaluation and Performance Comparison** (15 marks)